

# 23BMO1

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## Problem

Find all functions  $f: \mathbb{R} \rightarrow \mathbb{R}$  such that for all  $x, y \in \mathbb{R}$ ,

$$xf(x + f(y)) = (y - x)f(f(x)).$$

**Solution.** Let  $P(x, y)$  be the assertion.

$f \equiv 0$  is obviously a solution, so say  $\exists x_0$  such that  $f(x_0) \neq 0$ .  $P(0, y) \implies f(f(0)) = 0$ .

$$P(x, x) \implies f(x + f(x)) = 0.$$

We claim that  $f$  is injective.

FTSOC, assume  $\exists a \neq b$  such that  $f(a) = f(b)$ .

$$P(x_0, a) \implies \frac{f(f(x_0))(a - x_0)}{x_0} = \frac{f(f(x_0))(b - x_0)}{x_0} \implies a - x_0 = b - x_0 \implies a = b.$$

So  $x + f(x) = f(0)$ , or,  $f(x) = f(0) - x$ .

As a result, all the functions of the form  $f(x) = c - x$  work where  $c \in \mathbb{R}$ .